

ECU



101

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2019/2020 FIRST SEMESTER
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ELECTRICAL
AND ELECTRONIC, AEROSPACE, MECHANICAL, BIOSYSTEMS, BIOMEDICAL,
PETROLEUM AND CIVIL ENGINEERING)
ECU 101: PHYSICS FOR ENGINEERS I

DATE: MONDAY, 9TH DECEMBER, 2019

TIME: 2.00 P.M. - 4.00 P.M.

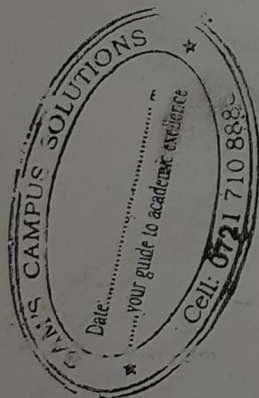
INSTRUCTIONS

1. Answer questions ONE and any other TWO questions.
2. The following constants may be useful:
 $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $g = 9.81 \text{ m/s}^2$ $R = 8.31 \text{ J/moleK}$
 Coefficient of linear expansion of steel $= 1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
 Young's modulus of steel is $= 2.1 \times 10^5 \text{ Nm}^{-2}$.
 Wien's constant $= 2.90 \times 10^{-3} \text{ m.K}$



QUESTION ONE.

- (a) (i) What is meant by work energy theorem? (1 mark)
- (ii) A 6-kg block initially at rest is pulled along a horizontal smooth surface by a constant force of 12 N. Find the speed of the block after it moves a distance of 3 m. (3 marks)
- (b) (i) What is meant by conservative force? (1 mark)
- (ii) Show that the force due to gravity is conservative (3 marks)
- (c) (i) What is meant by an inelastic collision? (1 mark)
- (ii) A 5 g bullet is fired into a 1 kg block of wood suspended from some light wires. The bullet is stopped by the block and the entire system swings through a height of 5 cm. If the collision is perfectly in elastic find the initial speed of the bullet. (3 marks)
- (d) (i) What is meant by viscosity of a liquid? (1 mark)
- (ii) A metal sphere of radius 0.5 cm and density 7874 kg/m^3 falls with a terminal velocity of 2 m/s in a fluid of density 920 kg/m^3 . Find the viscosity of the fluid. (3 marks)

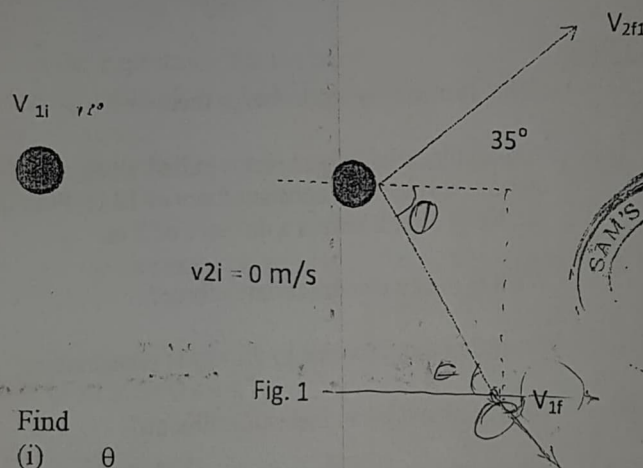


ECU 101

- (e) (i) What is meant by shear modulus? (1 mark)
- (ii) A block of rubber of height 5 cm and area 150 cm^2 is sheared with a force of 5 N. If a displacement of 1 cm is recorded find the shear modulus. (3 marks)
- (f) (i) Define the ~~elastic constant~~ ^{Poisson's} Poisson's ratio? (1 mark)
- (ii) A metal rod of length 20 cm is stretched to 25 cm. If it changes from 5 cm to 2 cm, find its Poisson's ratio. (3 marks)
- (g) (i) What is meant by thermal expansivity? (1 mark)
- (ii) A metal rod of length 20 cm is heated from 25°C to 80°C . If the change in length is 2 cm find its linear expansivity. (3 marks)
- (h) Differentiate between adiabatic and isochoric processes. (2 marks)

QUESTION TWO

- (a) In a game of billiards, a player wishes to "sink" the target ball in the corner pocket as shown in Fig. 1.



Find

- (i) θ (4 marks)
- (ii) v_{1f} if v_{1i} is 150 m/s. (4 marks)
- (iii) v_{2f} if v_{1i} is 150 m/s. (4 marks)
- (b) Two springs of spring constant k_1 and k_2 are joined end to end. One end is connected to a mass m and the other to a fixed wall. Find the expression for the frequency in Hz of the motion. (4 marks)
- (c) The springs above are now attached to m and their other ends to fixed walls. Find the expression for the frequency in Hz of the motion. (4 marks)

QUESTION THREE

- (a) Show that for an ideal gas in a quasi-static adiabatic process
- $PV^\gamma = \text{constant}$. (4 marks)
 - $TV^{\gamma-1} = \text{constant}$. (4 marks)
 - $TP^{(1-\gamma)/\gamma} = \text{constant}$. (4 marks)
- (b) A horizontal spring is found to be stretched 7.5 cm from its equilibrium position when a force of 0.34 N acts on it. Then a 1.125 kg body is attached to the end of the spring and is pulled 10 cm along the horizontal frictionless table from the equilibrium position. What is the
- force exerted by the spring on the 1.125 kg body just before it is released? (2 marks)
 - period of oscillation after release. (2 marks)
 - maximum speed of the vibrating body (2 marks)
 - maximum acceleration of the body (2 marks)

QUESTION FOUR

- (a) Below is a sketch of a truncated cross-section of a liquid in glass thermometer. Through heat energy transfer processes, explain clearly how the parts marked i). to v). contribute to accurate temperature measurement. (5 marks).
- (b) (i). Briefly explain how this type of thermometer is calibrated. (2 marks).
 (ii). List two suitable liquids for use in the thermometer, qualifying the properties contributing to the liquids suitability as thermometric substance. (2marks)

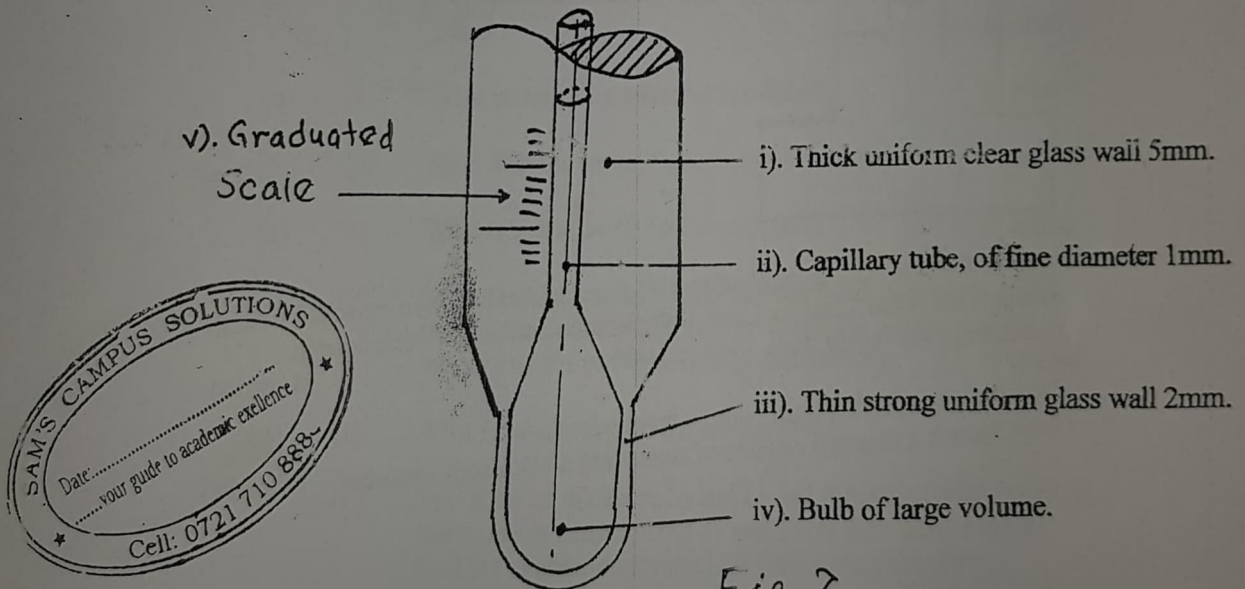


Fig. 2

- (c). Define the following mechanical properties of matter and in each case, mention one area in engineering field, where the properties find use.
- Elasticity. (ii). Plasticity. (iii). Defects. (4 marks)

- (d). Use a sketch of a unit cube undergoing tensile stress (σ) or otherwise to derive the elastic relation provided below, $K = \frac{Y}{3(1-2\nu)}$
 Where; Y, represent the Young's modulus, K, represents bulk modulus and
 ν , represents Poisson's ratio. (6 marks)

QUESTION FIVE

- (a). Rotation of a crane arm about a point O in the diagram below is defined by
 $\phi = 0.15t^2$.

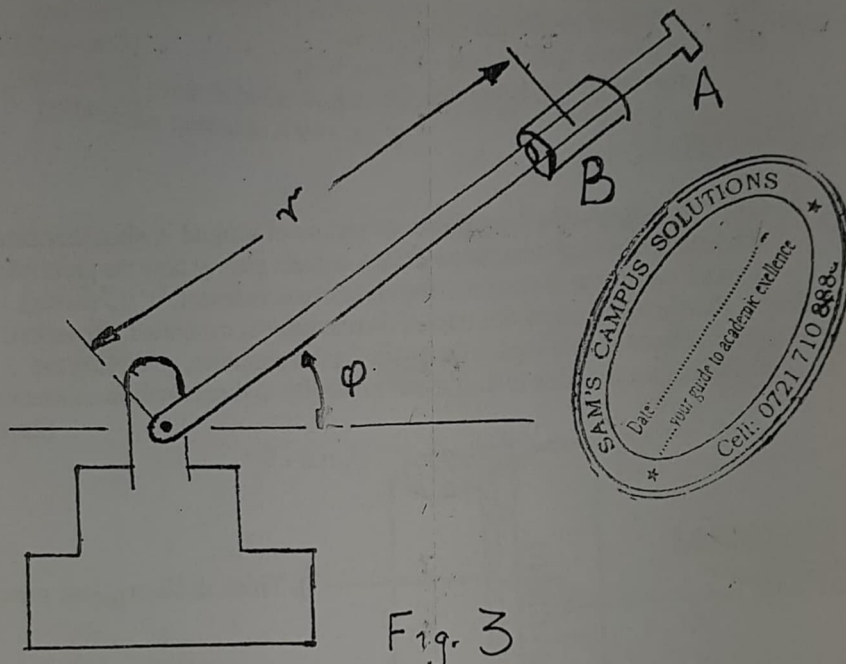
Where; ϕ , is in radians and t is in seconds.

Collar B slides along the arm such that $r = 0.9 - 0.12t^2$

Where; r , is in meters.

After the arm has rotated through 30° , determine the following;

- Total velocity of the collar. (4 marks)
- Total acceleration of the collar. (4 marks)
- Relative acceleration of the collar with respect to the arm. (3 marks)

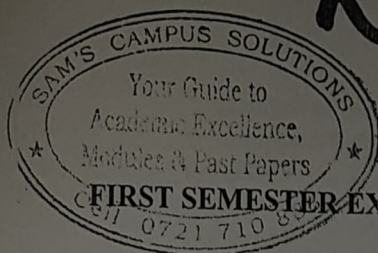


- (b). A force of 45N is required to hold a spring that has been stretched from its natural length of 12cm to 15cm. Calculate the amount of work done in stretching the spring from 15cm to 18cm. (4 marks)

- (c). Define a perfect blackbody in thermodynamics. (1 mark)
 Sketch a diagram capable of simulating a blackbody absorber. (4 marks)

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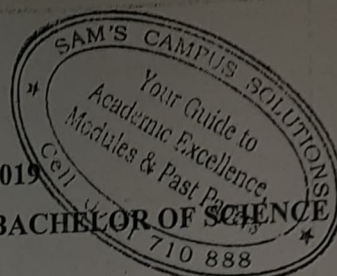
ECU 101



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE



ECU 101: PHYSICS FOR ENGINEERS I

DATE: Tuesday 11th December 2018

TIME: 4.30 p.m. – 6.30 p.m.

INSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.
2. The following constants may be useful:
 $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $g = 9.81 \text{ m/s}^2$ $R = 8.31 \text{ J/moleK}$
Coefficient of linear expansion of steel $= 1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
Young's modulus of steel is $= 2.1 \times 10^6 \text{ Nm}^{-2}$.
Wien's constant $= 2.90 \times 10^{-3} \text{ m.K}$

- Q1. (a) (i) Define work in vector notation. (1 mark)
- (ii) A particle undergoes a displacement $\vec{s} = (2\hat{x} + 3\hat{y}) \text{ m}$ while a constant force given by $\vec{F} = (5\hat{x} + 2\hat{y}) \text{ N}$ acts on the particle. Calculate the work done by the force and the angle between $\vec{s}$ and $\vec{F}$. (3 marks)
- (b) (i) What is meant by conservative force? (1 mark)
- (ii) Show that the spring force is conservative (3 marks)
- (c) (i) What is meant by an elastic collision? (1 mark)
- (ii) Show that for an elastic collision the relative velocity of the two masses before the collision equals the negative of the velocity of the two masses after the collision. (3 marks)
- (d) (i) What is meant by thermal conductivity? (1 mark)
- (ii) Two slabs of thickness L_1 and L_2 and thermal conductivities k_1 and k_2 are at temperatures T_1 and T_2 on their outer surfaces. Find the expression for the heat flow across the slabs. (3 marks)

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- (e) (i) What is meant by Stephan's law? (1 mark)
- (ii) The average radiated power from the sun reaching the earth's surface is 1.395 kW/m^2 . Calculate the absolute temperature of the sun's surface. (3 marks)
- (f) (i) What is meant by the kinetic theory of gases? (1 mark)
- (ii) A cylinder contains 3 moles of helium gas at a temperature of 400 K. How much heat must be transferred to the gas to increase its temperature to 600 K if the gas is heated at constant volume and C_v for helium is 12.5 J/molK ? (3 marks)
- (g) (i) What is meant by Stoke's law? (1 mark)
- (ii) A sphere of radius 0.2 cm drops in oil at 20 cm/s. If the viscosity of oil is 0.6055 Ns/m^2 , find the drag force. (3 marks)
- (h) (i) In words, what is meant by Reynold's number? (1 mark)
- (ii) Water is flowing in a 2 cm diameter pipe at 15 m/s. If the density and viscosity is 1000 kg/m^3 and 1.005 Ns/m^2 what is the Reynold's number? (3 marks)
- (iii) Differentiate between shear modulus and Poisson's ratio. (2 marks)

- Q2. (a) When a system is taken from state i to state f along the path iaf , it is found that $Q = 5 \text{ cal}$ and $W = 20 \text{ cal}$. Along the path ibf $Q = 36 \text{ cal}$ as shown in Fig. 1.

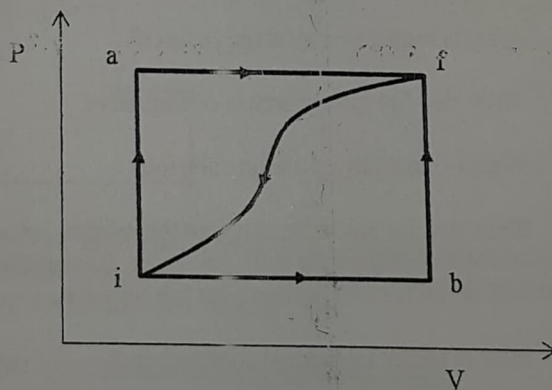


Fig. 1

If $Q_{ibf} = 36 \text{ cal}$, $Q_{iaf} = 50 \text{ cal}$ and $W_{iaf} = 20 \text{ cal}$, find

- (i) W_{ibf} (2 marks)
- (ii) Q_{fi} assuming $W_{fi} = -13 \text{ cal}$ (2 marks)
- (iii) U_f assuming $U_i = 10 \text{ cal}$ (2 marks)



- (iv) Q_{ib} assuming $U_b = 22$ cal (2 marks)
 (v) Q_{bf} assuming $U_b = 22$ cal (2 marks)
- (b) Find the x- and y-coordinates of the centre of mass of a circular sheet of radius a . (4 marks)
- (c) (i) A steel elevator cable breaking stress is 7.2×10^8 Pa and is made to support a 2000 kg elevator whose maximum upward acceleration is 2 m/s^2 . What diameter cable should be used if the maximum stress is one quarter the breaking stress? (3 marks)
- (ii) A copper wire of area 0.05 cm^2 and length 5 cm is attached end to end to a steel wire of length 3 m and area 0.02 cm^2 . If the wires are stretched to a tension of 200 N find the change in length of the combination. Assume $Y_s = 2 \times 10^{11}$ Pa and $Y_c = 1.1 \times 10^{11}$ Pa. (3 marks)

- Q3. (a) Show that for an ideal gas in a quasistatic adiabatic process (4 marks)
 (i) $PV^\gamma = \text{constant}$. (3 marks)
 (ii) $TV^{\gamma-1} = \text{constant}$. (3 marks)
 (iii) $TP^{(1-\gamma)/\gamma} = \text{constant}$.

- (b) Using kinetic theory show that the
 (i) total pressure of molecules against cube walls is given by

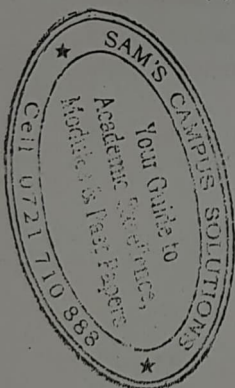
$$P = \frac{Nm\langle v^2 \rangle}{3V} \quad (4 \text{ marks})$$

- (ii) internal energy molecules is given by

$$U = \frac{Nm\langle v^2 \rangle}{2} \quad (3 \text{ marks})$$

- (iii) temperature of the system is given by

$$T = \frac{2N\langle K \rangle}{3nR} \quad (3 \text{ marks})$$



- Q4. (a) What is the thermometric property used in each of the following thermometers to measure temperature? (2 marks)
 (i) Liquid in glass thermometer. (2 marks)
 (ii) Thermistor thermometer. (2 marks)
 (iii) Pyranometer thermometer.
- (b) Give one example each of the following thermodynamic processes; (2 marks)
 (i) An adiabatic change (2 marks)
 (ii) An isothermal change.
- (c) (i) Explain the use of vectors in the field of engineering? (1 mark)

(ii) If A and B are two vectors such that $A=4i+6j-3k$ and $B=-2i-5j+7k$, determine their dot product. Also, determine their cross product. (6 marks)

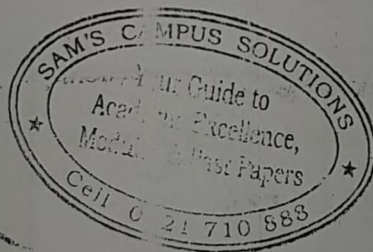
(iii) What are some significant applications of elastic constants? (3 mark)

Q5. (a) (i) What influences elastic property in a material? (2 marks)
(ii) With the help of a well labelled graph, describe the behaviour of a ductile wire under weight of an increasing load. (6 marks)
(iii) Calculate the new dimension of a metre-long, 3mm diameter steel wire due to a temperature change from 24°C to 32°C . (4 marks)

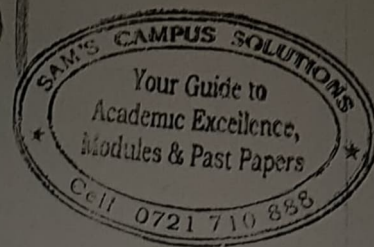
(b) Show that the:

- (i) volume expansivity is thrice that of linear expansivity. (4 marks)
(ii) area expansivity is twice that of linear expansivity. (4 marks)

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ECU 101



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(ELECTRICAL AND ELECTRONIC, AEROSPACE, MECHANICAL, BIOSYSTEMS,
BIOMEDICAL PETROLEUM AND CIVIL ENGINEERING)

ECU 101: PHYSICS FOR ENGINEERS I

DATE: Tuesday, 6th December, 2016

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.
2. The following constants may be useful:
 $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ $g = 9.81 \text{ m/s}^2$ $R = 8.31 \text{ J/moleK}$
Coefficient of linear expansion of steel $= 1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
Young's modulus of steel is $= 2.1 \times 10^6 \text{ Nm}^{-2}$.
Wien's constant $= 2.90 \times 10^{-3} \text{ m.K}$

- Q1. (a) (i) What is the difference between the centre of mass of an object and the centre of gravity of the same object? (1 mark)
- (ii) Show that the centre of mass of a uniform rod of mass M , length L and linear density $\lambda = M/L$ lies midway between its ends. (3 marks)
- (b) (i) How are vectors significant in the field of engineering? (1 mark)
- (ii) If A and B are two vectors such that $A=4i+6j-3k$ and $B=-2i-5j+7k$, determine the direction cosines and the angle between A and B . (3 marks)
- (c) (i) In words, what is meant by a conservative force? (1 mark)
- (ii) Show that force due to a spring is conservative. (3 marks)
- (d) (i) In words, what is meant by first law of thermodynamics. (1 mark)
- (ii) Calculate the maximum wavelength of radiation emitted by a blackbody at 100°C ? (3 marks)

- (e) (i) What is meant by the kinetic theory of gases? (1 mark)
- (ii) A cylinder contains 3 moles of helium gas at a temperature of 400 K. How much heat must be transferred to the gas to increase its temperature to 600 K, if the gas is heated at constant volume and C_v for helium is 12.5 J/mol K ? (3 marks)
- (f) (i) In words, what is meant by bulk modulus. (1 mark)
- (ii) A material with shear modulus $2.5 \times 10^{10} \text{ N/m}^2$ and height 3 cm undergoes a shear distance of 2 mm. Find the applied force on area 15 cm^2 . (3 marks)
- (g) (i) In words, what is meant by Reynold's number? (1 mark)
- (ii) Water is flowing in a 2 cm diameter pipe at 15 m/s. If the density and viscosity is 1000 kg/m^3 and 1.005 N s/m^2 . what is the Reynold's number? (3 marks)
- (h) Differentiate between Reynold's number and Poisson's ratio. (2 marks)
- Q2. (a) Show that the:
- i. volume expansivity is thrice that of linear expansivity. (4 marks)
- (ii) area expansivity is twice that of linear expansivity. (4 marks)
- (b) With the help of a graph, show that the work done by a closed system during an isothermal process is given by $W = RT \log_e [V_2/V_1]$. Where the symbols represent their usual values (4 marks)
- (c) A pendulum consists of a sphere of mass 100 g attached to a light cord of length 1 m. The sphere is released from rest when the cord makes an angle 40° with the vertical. Determine the
- (i) speed of the sphere when it is at its lowest point. (4 marks)
- (ii) tension in the cord when the sphere is at its lowest point (4 marks)
- Q3. (a) A reversible heat engine carries 1 mole of an ideal monoatomic gas around the cycle shown in Fig 1. Process 1 to 2 isochoric; Process 2 to 3 is adiabatic Process 3 to 1 is isobaric $T_1 = 300 \text{ K}$, $T_2 = 600 \text{ K}$, $T_3 = 455 \text{ K}$

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6th Dec 2016

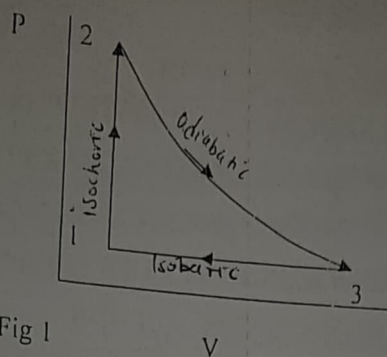


Fig 1

Calculate

- (i) ΔU_{12}
- (ii) W_{23}
- (iii) Q_{31}

(4 marks)

(4 marks)

(4 marks)

- (b) Study the frictionless system shown in Fig. 2, where $m_1 = 2$ kg, $m_2 = 3$ kg, $m_3 = 10$ kg and $\theta = 30^\circ$.

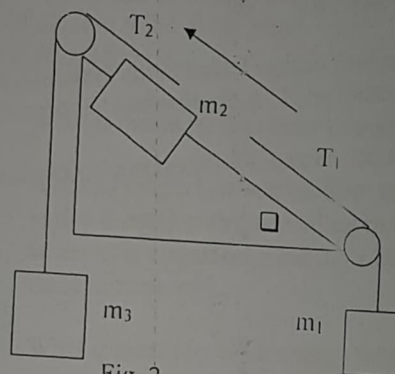
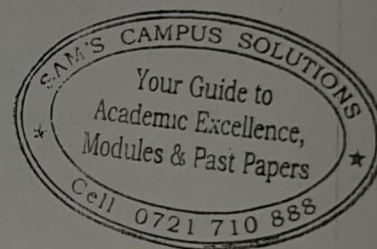


Fig. 2

- (i) Find the acceleration
- (ii) Find tensions T_1 and T_2

(4 marks)

(4 marks)



- Q4. A horizontal spring is found to be stretched 7.5 cm from its equilibrium position when a force of 0.34 N acts on it. Then a 1.125 kg body is attached to the end of the spring and is pulled 10 cm along the horizontal frictionless table from the equilibrium position. What is the

- (i) force exerted by the spring on the 1.125 kg body just before it is released? (3 marks)
- (ii) period of oscillation after release. (3 marks)
- (iii) maximum speed of the vibrating body (3 marks)

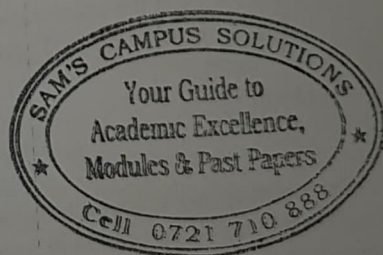
- (iv) maximum acceleration of the body (3 marks)
- (v) total energy of the oscillating system (3 marks)
- (vi) equation of the displacement of the body as a function of time (5 marks)

- Q5. (a) State three important postulates of the kinetic theory of gases. (3 marks)
- (c) Calculate the root mean-square speed of a hydrogen molecule at a temperature of 300K. [Mass of hydrogen atom is $1.67 \times 10^{-27} \text{kg}$.] (4 marks)
- (b) A process called gaseous diffusion is sometimes used to separate isotopes of Uranium -235 and Uranium -238, (^{235}U and ^{238}U). Use your knowledge to figure out how this is achieved. (3 marks)
- (c) Use the first law of thermodynamics to show that the specific heat capacity of a gas at constant volume (C_v) can be expressed as;

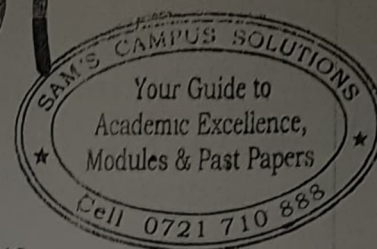
$$\left(\frac{\Delta Q}{dT} \right)_v = \left(\frac{\partial U}{\partial T} \right)_v = C_v$$

Where; Q, is the heat energy added to the system U, is the internal energy of the system and T, is the absolute temperature. (10 marks)

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UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
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BIOMEDICAL PETROLEUM AND CIVIL ENGINEERING)

ECU 101: PHYSICS FOR ENGINEERS I

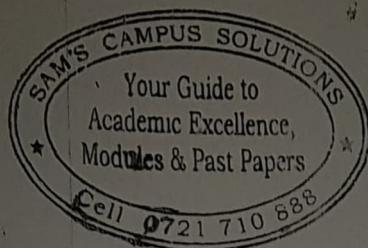
DATE: TUESDAY 20TH FEBRUARY 2018

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS

1. Answer questions ONE and any other TWO questions.
2. The following constants may be useful:
 $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ $g = 9.81 \text{ m/s}^2$ $R = 8.31 \text{ J/moleK}$
Coefficient of linear expansion of steel = $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
Young's modulus of steel is = $2.1 \times 10^6 \text{ Nm}^{-2}$.
Wien's constant = $2.90 \times 10^{-3} \text{ m.K}$

- Q1. (a) (i) What is meant by the center of mass of an object? (1 mark)
- (ii) A rod of length L is non-uniform and the mass per unit length varies linearly with x according to the expression $\lambda = \alpha x$, where α is a constant. Find the x coordinate of the center of mass in terms of L . (3 marks)
- (b) (i) What is meant by work-energy theorem? (1 mark)
- (ii) A 6 kg block initially at rest is pulled to the right along a horizontal smooth surface by a constant horizontal force of 12 N. Find the speed of the block after it moves a distance of 3 m. (3 marks)
- (c) (i) What is meant by inelastic collision? (1 mark)
- (ii) A 5 g bullet is fired into a 1 kg block of wood suspended from some two light wires. The bullet is stopped by the block and the entire system swings to a height of 5 cm. If the collision is perfectly inelastic find the initial speed of the bullet. (3 marks)
- (d) (i) What is meant by viscosity? (1 mark)
- (ii) A metal plate of area 0.05 m^2 is connected to an 8-g mass via



string that passes over a massless-frictionless pulley. A lubricant with a film thickness of 0.3 mm is placed between the plate and the surface. When released the plate moves with a speed of 0.085 m/s. Find the viscosity of the lubricant.

(3 marks)

- (e) (i) What is meant by first Stephan's law? (1 mark)
- (ii) The average radiated power from the sun reaching the earth's surface is 1.395 kW/m^2 . Calculate the absolute temperature of the sun's surface. (3 marks)
- (f) (i) What is meant by the kinetic theory of gases? (1 mark)
- (ii) A cylinder contains 3 moles of helium gas at a temperature of 400 K. How much heat must be transferred to the gas to increase its temperature to 600 K if the gas is heated at constant volume and C_v for helium is 12.5 J/molK ? (3 marks)
- (g) (i) In words, what is meant by Reynold's number? (1 mark)
- (ii) Water is flowing in a 2 cm diameter pipe at 15 m/s. If the density and viscosity is 1000 kg/m^3 and 1.005 Ns/m^2 what is the Reynold's number? (3 marks)
- (iii) Differentiate between shear modulus and Poisson's ratio. (2 marks)

- Q2. (a) Show that the
- (i) Volume expansivity is thrice that of linear expansivity. (4 marks)
- (ii) Area expansivity is twice that of linear expansivity. (4 marks)
- (b) Find the x- and y-coordinates of the centre of mass of a ^{quarter} circular sheet of radius a . (4 marks)
- (c) An 1800 kg car stopped at a traffic light is struck from the rear by another car of mass 900 kg moving at 20 m/s. If the two cars get entangled, what is
- (i) their common velocity. (4 marks)
- (ii) the loss of kinetic energy. (4 marks)

- Q3. (a) Show that for an ideal gas in a quasi-static adiabatic process
- (i) $PV^\gamma = \text{constant}$. (4 marks)
 - (ii) $TV^{\gamma-1} = \text{constant}$. (4 marks)
 - (iii) $TP^{(1-\gamma)/\gamma} = \text{constant}$. (4 marks)

- (b) Fig. 2 shows a crane used in a building construction. The mass of the movable 18 m boom is 1000 kg and the mass of the base is 10,000 kg.

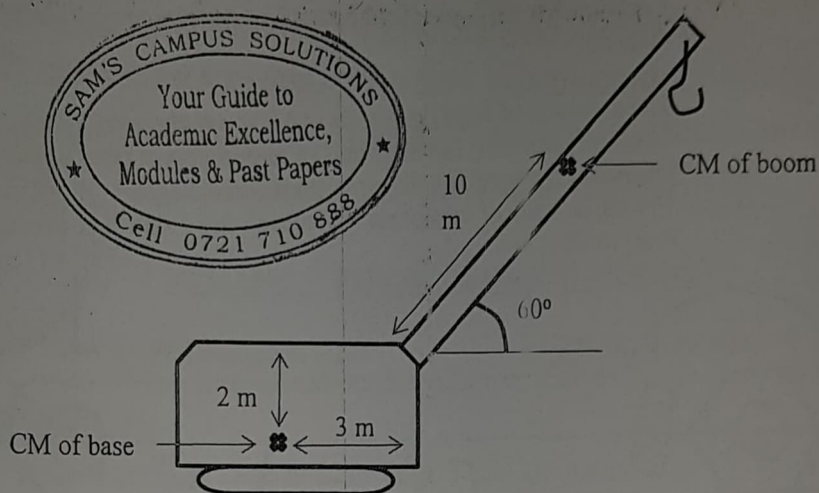
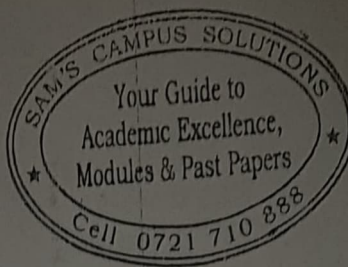


Fig. 1

- (i) Find the x and y coordinates of the center of mass. (4 marks)
 - (ii) How large a mass can be hung from the end of the boom before it will topple? (4 marks)
- Q4. (a) (i). Explain the necessary steps for establishing the upper fixed and lower fixed points of a liquid in glass thermometer. (4 marks)
- (ii). How then would you calibrate the scale to read degree Celsius? (1 mark)
- (b) A nut is tightened by applying a 45N force on a 0.30m wrench spanner, as shown in the figure 2.
- (i). Calculate the magnitude of the torque about the nut's centre. (4 marks)
 - (ii). Calculate the torque vector if the nut is right threaded. (1 mark)
- (c) (i). Use a well labelled graph to describe the behaviour of a ductile wire under the influence of an ever increasing load. (6 marks)
- (ii). A 2 mm diameter steel wire is stretched between two fixed points at 30° C. Determine the dimensional changes expected when temperature falls to 20° C? (4 marks)



- Q5. (a). (i). Define impulse phenomenon in terms of force (f) and time (t). (1 mark)
- (ii). Explain scientifically the action of air-bags in standard automobiles in the unfortunate event of a head-on collision. (5 marks)
- (b). (i). Define momentum of a body. (2 marks)
- (ii). Explain what you understand by the impulse-momentum change theorem. (3 marks)
- (c). Three vectors of magnitude a , $2a$ and $3a$ meet at a point. Their directions are along the diagonals of three adjacent faces of a cube.
- (i). Represent the vectors in terms of forces on a diagram. (2 marks)
- (ii). Use the concept of resolution of forces to determine their resultant magnitude. (7 marks)

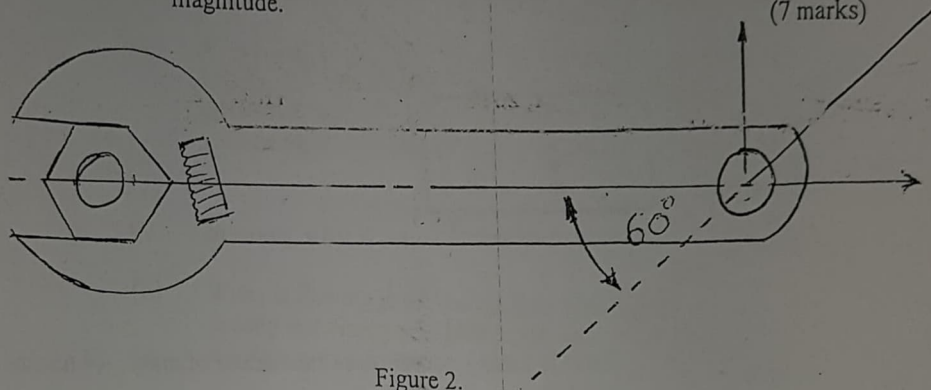


Figure 2.

Deriving Impulse-Momentum Theorem

$$F = ma$$

$$F = m \frac{dv}{dt}$$

$$F = \frac{dP}{dt}$$

$$dP = F dt$$

$$\int_{P_0}^P dP = F \int_{t_0}^t dt \Rightarrow [P]_{P_0}^P = F \int_{t_0}^t dt$$

$$P - P_0 = F \int_{t_0}^t dt \Rightarrow P - P_0 = \Delta P$$

$$\Delta P = F \int_{t_0}^t dt \Rightarrow \Delta P = F(t - t_0)$$

$$\text{Let } t - t_0 = \Delta t \Rightarrow \Delta P = F \Delta t$$

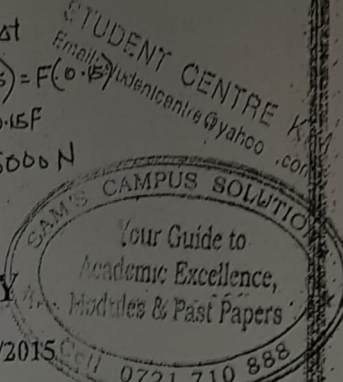
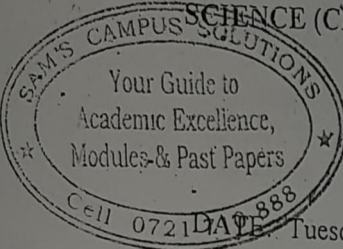


KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (CIVIL/MECHANICAL/ELECTRICAL/BIO MEDICAL/BIOSYSTEMS
AND PETROLEUM ENGINEERING)

ECU 101: PHYSICS FOR ENGINEERS I



Tuesday, 9th December, 2014

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

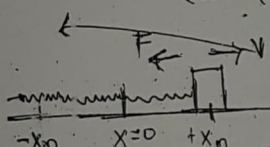
1. Answer question ONE and any other TWO questions.
2. The following constants and table may be useful:

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2 \quad g = 9.81 \text{ m/s}^2 \quad R = 8.31 \text{ J/mol K}$$

Q1. (a) (i) In words, explain what is meant by a conservative force?

Force that does no work round a close path

$$(1 \text{ mark}) = \frac{1}{2}mv^2 = \frac{1}{2}mv^2$$



(ii) Show that force due to a spring is conservative. (3 marks)

$$F = -kx$$

Work done by spring from $x_i = -x_m$ to $x_f = 0$

$$W = \int_{x_i}^{x_f} F dx = \int_{-x_m}^0 (-kx) dx$$

$$W = \frac{1}{2}kx^2$$

Work done from $x_i = 0$ to $x_f = x_m$

$$W = \int_0^{x_m} (-kx) dx = -\frac{1}{2}kx^2$$

$$\text{Net work} = \frac{1}{2}kx^2 + -\frac{1}{2}kx^2 = 0$$

(i) In words, explain what is meant by work energy theorem?

Work W done by the resultant force acting on a body is equal to the change in the kinetic energy of the body.

$$W = K - K_0 = \Delta K$$

(ii) A 6-kg block initially at rest is pulled to the right along a horizontal smooth surface by a constant, horizontal force of 12 N. Find the speed of the block after it moves a distance of 3 m. (3 marks)

Constant Force

$$a = \frac{F}{m}$$

$$x = \frac{v^2 - v_0^2}{2a}$$

$$W = F \Delta x$$

$$= m \frac{(v - v_0)(v + v_0)}{2}$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

$$\text{Let } K = \frac{1}{2}mv^2$$

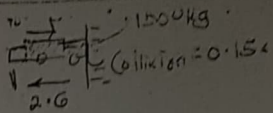
$$\text{Thus } K = K - K_0 = \Delta K$$

(i) In words, what is meant by impulse-momentum theorem?

Impulse of force F equals to change in kinetic momentum of the particle.

$$I = \Delta P = F \Delta t$$

(ii) An automobile of mass 1500 kg collides with a wall. The initial and final velocities of the automobile are -15.0 m/s and 2.6 m/s, respectively. If the collision lasts for 0.15 s, find the average force exerted on the automobile. (3 marks)



$$\Delta P = F \Delta t$$

$$m(v - v_0) = F \Delta t$$

$$1500(2.6 - -15) = F(0.15)$$

$$26400 = 0.15F$$

$$F = 176000 \text{ N}$$

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

(ii) From Work-Energy Theorem:

$$K = K - K_0$$

$$12 \times 3 = \frac{1}{2} \times 6 \times v^2$$

$$F_x = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

$$12 = v^2$$

$$v = \sqrt{12} = 3.46 \text{ m/s}$$

$$\text{Let } \frac{1}{2}mv_0^2 = 0$$

- (d) (i) In words, what is meant by first law of thermodynamics. (1 mark)
- (ii) Show that the work done by a system depends not only on the final states but also on the path taken. (3 marks)
- (e) (i) What is meant by the kinetic theory of gases? (1 mark)
- (ii) One mole of helium gas in a large volume is cooled to 31.5 K. What is the average value of the x-component of velocity squared $\langle v_x^2 \rangle$? Mass of one mole of helium = 6.7×10^{-27} kg. (3 marks)
- (f) (i) In words, what is meant by bulk modulus. (1 mark)
- (ii) A body of volume 0.5 m^3 undergoes a volume contraction of 0.03 m^3 when subjected to a pressure change of $5 \times 10^5 \text{ N/m}^2$. Find its bulk modulus. (3 marks)
- (g) (i) In words, what is meant by viscosity of a fluid? (1 mark)
- (ii) A metal sphere of radius 1 cm and density 1500 kg/m^3 is dropped in an oil column of density 700 kg/m^3 . If it attains a terminal velocity of 9.2 cm/s, find the viscosity of oil. (3 marks)

(h) Differentiate between Bernoulli's principle and equation of continuity. (2 marks)

Q2. (a) A 0.5 kg mass of putty is hurled at 1 m/s against a 0.5 kg mass attached to a spring as shown in Fig. 1. The mass attached to the

Masses A and B
Cons Momentum
 $5 \times 1 + (0.5 \times 0) = (0.5 + 0.5) v$
 $0.5 = 1v$
 $v = 0.5 \text{ m/s}$
Kinetic energy after collision
 $\frac{1}{2} (m_A + m_B) v_f^2$
 $= \frac{1}{2} (1) (0.5)^2$
 $= 0.125 \text{ J}$

$$m_A v_{A1} + m_B v_{B1} = (m_A + m_B) v_f$$

$$W_s = \frac{1}{2} k x^2$$

$$W_s = K - K_0$$

$$W_s = \frac{1}{2} m v^2 - \frac{1}{2} m v_0^2$$

0.5 kg 1 m/s 0.5 kg k

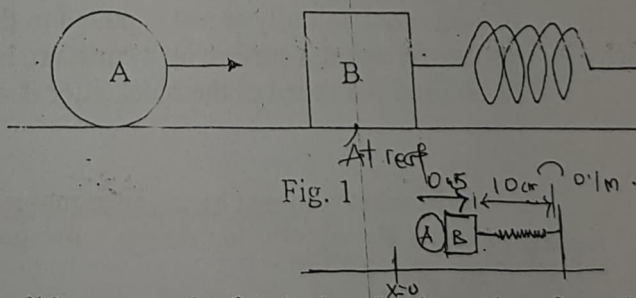


Fig. 1

$$\frac{1}{2} k x^2 = \frac{1}{2} m v^2$$

$$\frac{1}{2} k (0.1)^2 = 0.125$$

$$k = \frac{2 \times 0.125}{0.1^2}$$

$$= 25 \text{ N/m}$$

Final KE = $\frac{1}{2} m_A v_A^2$

$$0.5 \times 1^2 = 0.125 + \frac{1}{2} k x^2$$

$$0.25 - 0.125 = \frac{1}{2} k x^2$$

$$\frac{0.125 \times 2}{0.1^2} = k = 25 \text{ N/m}$$

Spring slides across the frictionless horizontal surface, depressing the spring a maximum distance of 10 cm. Find the spring constant if the putty sticks to the mass on the spring. (7 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

more m_1
 m_2 is init
 A_1

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

ECU 101

9th DEC 2014

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 Email: studentcentre@kmm.ac.lk

- (b) A 135 kg defensive lineman traveling at 5 m/s makes a completely inelastic collision with an 85 kg quarterback at rest. They are observed to slide 2 m on a wet grass. What is the
- coefficient of friction between the grass and the players? (7 marks)
 - energy lost in the collision? (6 marks)

Q3. (a) An ideal monatomic gas is taken from point 1 on the T_1 isotherm to point 3 on the T_2 isotherm along the path 1-2-3 shown in Fig. 2.

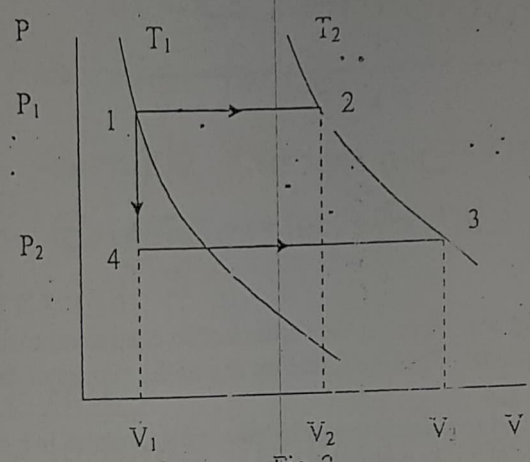
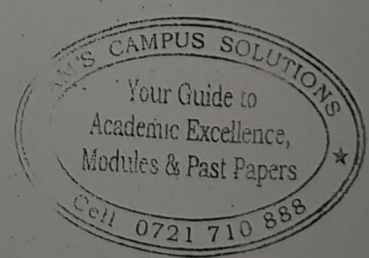


Fig. 2

- In terms of the parameters shown, calculate the change in internal energy of the gas and the heat added to it in this process (10 marks)
- Suppose the path 1-4-3 is followed instead. Calculate the heat added along this path. (3 marks)



Consider a 2d problem

m₂

but glancing A

Q4b
- direction
 $\Rightarrow \frac{mv^2}{r} = T \cos \theta + T' \cos \theta \quad (1)$

- direction
 $\Rightarrow T \sin \theta = T' \sin \theta + mg \quad (2)$
dividing eq 2 by eq 1

$\frac{v^2}{r} \tan \theta = T \cos \theta \tan \theta + T' \cos \theta \tan \theta$
 $\frac{v^2}{r} \tan \theta = T \sin \theta + T' \sin \theta \quad (3)$

subtracting eq 1 from 2

$\sin \theta - \frac{mv^2}{r} \tan \theta = mg - T \sin \theta$

$\sin \theta + T \sin \theta = mg + \frac{mv^2 \tan \theta}{r}$

$T = \frac{mg + \frac{mv^2 \tan \theta}{r}}{\sin \theta}$

$T = \frac{mv^2}{2r \cos \theta} + \frac{mg}{2 \sin \theta} \quad (4)$

Substituting eq 4 in 1

$\frac{v^2}{r} = \left(\frac{mv^2}{2r \cos \theta} + \frac{mg}{2 \sin \theta} \right) (\cos \theta + T' \cos \theta)$

$\frac{mv^2}{r} = \frac{mv^2}{2r} + \frac{mg}{2 \tan \theta}$
 $T' \cos \theta = \frac{mv^2}{2r} - \frac{mg}{2 \tan \theta}$

$T' = \frac{mv^2}{2r \cos \theta} - \frac{mg}{2 \sin \theta}$

$\frac{mv^2}{2r} - \frac{mg}{2 \tan \theta} = T' = \frac{mv^2}{2r \cos \theta} - \frac{mg}{2 \sin \theta}$

For the thermodynamic cycle shown in Fig. 3, use the $TV^{\gamma-1} = \text{constant}$ and the ideal gas law to show that $V_2/V_1 = V_3/V_4$.

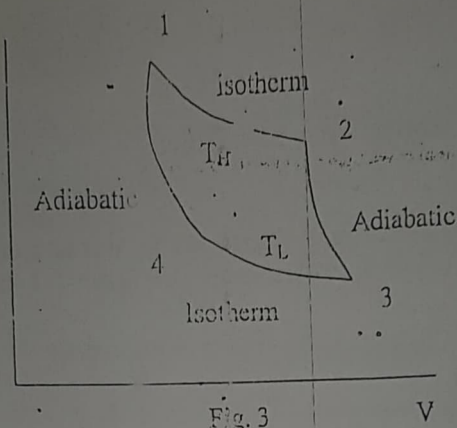


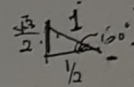
Fig. 3

$\theta = 60^\circ$

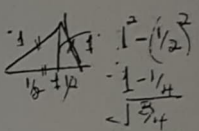
$\sin \theta = \frac{\sqrt{3}}{2}$

$\cos \theta = \frac{1}{2}$

LC, TCHT



(7 marks)



(11 marks)

Three 1000 kg masses are at the vertices of an equilateral triangle $r = 1$ m on its sides. Find the forces on any of the masses on the third.

A mass attached to a vertical post by two strings, rotates in a circle at constant speed v as shown in Fig 4.

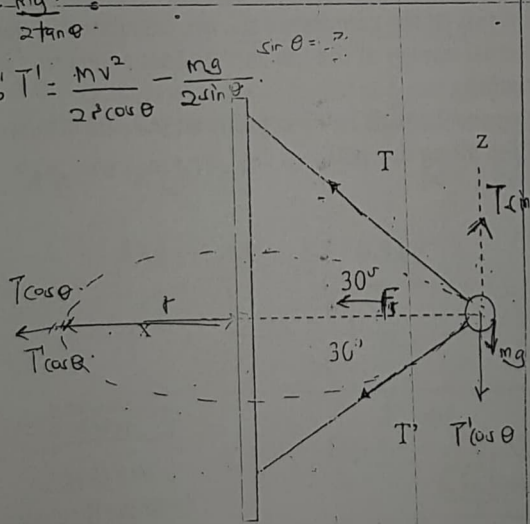
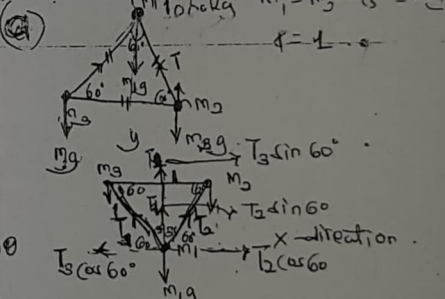


Fig. 4



$\sum F_x = T_2 \cos 60 - T_1 \cos 60 = 0$

$T_2 \cos 60 = T_1 \cos 60 \quad (1)$

$\sum F_y = T_1 \sin 60 + T_2 \sin 60 - mg = 0$

Eq 1: $T_2 = \frac{T_1 \cos 60}{\cos 60} = T_1$

Substituting for T_2 in eq 2

$T_1 \frac{\sqrt{3}}{2} + T_1 \frac{\sqrt{3}}{2} = 9.8 \times 1000$

$\sqrt{3} T_1 = 9800 \Rightarrow T_1 = \frac{9800}{\sqrt{3}} = 5658.03 \text{ N}$

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Circular Motion

$a_r = \frac{v^2}{r} \quad (1)$

$F_r = m a_r = \frac{mv^2}{r} \quad (2)$

$\frac{mv^2}{r} = T \sin \theta$

$T = \frac{2\pi r}{T} = \frac{2\pi r}{\sqrt{\frac{r}{g \tan \theta}}} \Rightarrow \frac{4\pi^2 r^2}{T^2} = \frac{r}{g \tan \theta}$
 $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{4\pi^2 r \sin \theta}{g \cos \theta}$

max m,
m₂ is in

DUE. 05-12-2018



SEMESTER I CAT 2 ECU 101 PHYSICS FOR ENGINEERS I

28-11-2018

18/19

1 Work: Differentiate scientifically between
i). Useful work and ii). Useless work

[8]

Force: Explain the following force concepts;
i). Frictional forces ii). Gravitational force
iii). Conservative forces

[9]

2. What is simple harmonic motion (SHM)?
Where in Engineering fields are SHM
encountered?

[8]

3. Fluid Flow:

a). Use diagrams to show laminar and turbulent
fluid flow.

What is the significance of Reynold's
number in fluid motion.

[10]

b). Analyse the given fluid flow equations
and briefly explain what parameter
is represented by each symbol.

is enough for
you to know the

$$i). \eta = \frac{F}{A \cdot \frac{dv}{dr}} \Rightarrow \eta = \frac{F}{A} \frac{dr}{dv} \text{ (Nm}^2\text{s)}$$

[8]

$$ii). N_R = \frac{\rho v D}{\eta} \quad iii). F = 6\pi\eta r v$$

$$iv). \frac{dV}{dt} = \frac{\pi R^4}{8\eta} \frac{P_1 - P_2}{L}$$

4. Explain the difference in solid & thermal (k) [7]
conductivities.

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$$\begin{array}{r} 1.5 \\ \times 58 \\ \hline 120 \\ 750 \\ \hline 87.0 \end{array}$$

(5 marks)

Q5. (a) A wedge car as shown in Fig. 5 is accelerating to the left. The coefficient of static friction between the block and the incline plane is μ . Find the maximum acceleration the wedge car can have before the block slides up the plane. (10 marks)

Smoot

Dividing eqt 1 by 2:

$$V^2 = Rg \tan \theta$$

$$V = \sqrt{Rg \tan \theta}$$

(i) rough	(6 marks)
(ii) smooth	(4 marks)

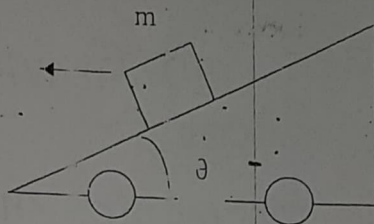
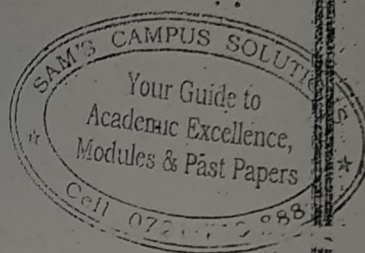
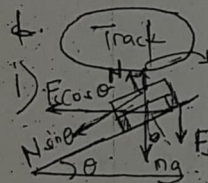


Fig 5



XXXXXXXXXXXXXXXXXXXXENDXXXXXXXXXXXXXXXXXXXX



$$N \cos \theta = mg + E \sin \theta \quad - (2)$$

But $F_s = \mu mg$.

Thus: Multiply eq. 1 by $\tan \theta$.

$$\frac{m v^2}{r} \tan \theta = N \cos \theta + F_s \sin \theta \quad \text{--- (3)}$$

Substituting N_{core} in eq. 3.

$$\frac{mv^2 \tan \theta}{D} = F \sin \theta + F \sin \theta + mg$$

$$\frac{mv^2 \tan \theta}{R} = 2F \sin \theta$$

subtracting eq. 3 from 2

$$N \cos \theta - \frac{mv^2 \tan \theta}{R} = mg - N \cos \theta$$

$$N = \frac{mg + \frac{mv^2 \tan \theta}{R}}{2 \cos \theta} ; N = \frac{mg}{2 \cos \theta} + \frac{mv^2}{2R \sin \theta} \quad (F)$$

Substituting eqt. 4 in 1.

$$\left(\frac{mg}{2\cos\theta} + \frac{mv^2}{2R\sin\theta} \right) \sin\theta + F_c \cos\theta = \frac{mv^2}{R}$$

$$\frac{mg \tan \theta}{2} + \frac{mv^2}{2R} + F_c \cos \theta = \frac{mv^2}{R}$$

$$\frac{mg \tan \theta}{2} + F_r \cos \theta = \frac{mv^0}{R} - \frac{mv^2}{R}$$

$$\frac{mg \tan \theta}{2} + F_c \cos \theta = \frac{mv^2}{2R} ; \text{ But } F_c = \mu mg$$

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Page 5 of 5

$$\frac{mg \tan \theta}{2} + \mu mg \cos \theta = \frac{mv^2}{2R}$$

$$\left(\frac{1}{2} \tan \theta + \cos \theta \right) \cancel{mg} = \frac{mv^2}{2R}$$

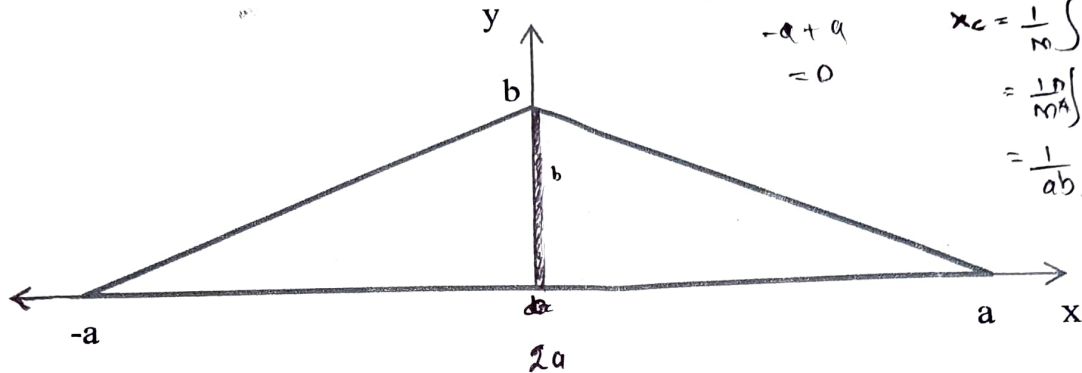
$$2^{\text{nd}} = \left(\frac{1}{2} \tan \theta \right) \text{ prove}$$

22

KENYATTA UNIVERSITY
ECU101 PHYSICS FOR ENGINEERS I TIME 60 MIN

$g = 9.8 \text{ m/s}^2$

Q1. A plate of mass M is shown below. Locate the center of mass (x_c and y_c) assuming the object has a uniform mass per unit area.

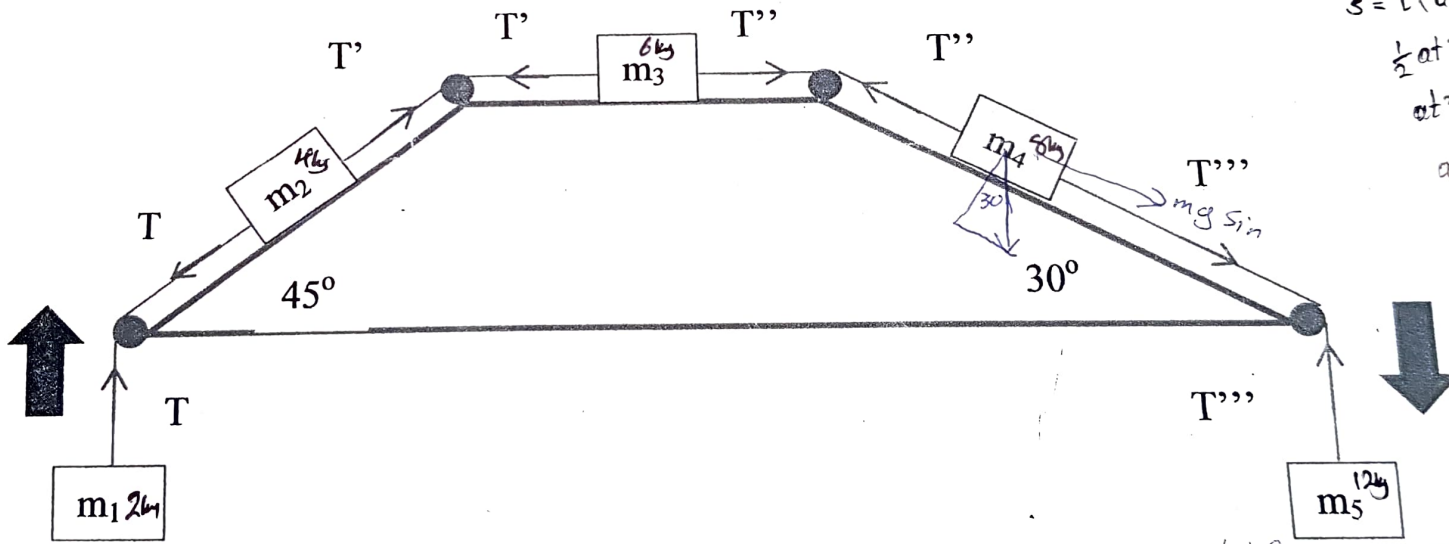


$x_c = \frac{1}{M} \int x dm$
 $dA = dx \times b$
 $dm = \sigma dA = \frac{M}{A} dA$
 $x_c = \frac{1}{M} \int x \sigma dA$
 $= \frac{1}{M} \int_{-a}^a x \sigma b dx$
 $= \frac{1}{M} \sigma b \int_{-a}^a x dx$
 $= \frac{1}{M} \sigma b \left[\frac{x^2}{2} \right]_{-a}^a$
 $= \frac{1}{M} \sigma b \left(\frac{a^2}{2} - \frac{(-a)^2}{2} \right)$
 $= \frac{1}{M} \sigma b (0) = 0$

$A = \frac{1}{2} \times a \times b$
 $A = ab$
 $\left(\frac{1}{2} \times a \times b \right)$
 $= \frac{1}{2} ab$
 $A = \frac{1}{2} ab$

Q2. An automobile travelling at 48 km/h can be stopped in a minimum distance of 40 m by applying the brakes. If the same automobile is travelling at 96 km/h, what is the minimum stopping distance? Use work energy theorem

Q3. Five masses are connected by light strings as shown below.



If $m_1 = 2 \text{ kg}$, $m_2 = 4 \text{ kg}$, $m_3 = 6 \text{ kg}$, $m_4 = 8 \text{ kg}$, $m_5 = 12 \text{ kg}$, find

- the acceleration
- T $mg = 2 \times 9.8 = 19.6 \text{ N}$
- T' $mg \sin \theta + T = 41.6 \text{ N}$
- T''
- T'''

$\text{Net } F = ma = T - W$
 $= ma - T$
 $F =$



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2020/2021
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ELECTRICAL AND ELECTRONICS, AEROSPACE, MECHANICAL,
BIOSYSTEMS, BIOMEDICAL PETROLEUM AND CIVIL ENGINEERING

ECU 101: PHYSICS FOR ENGINEERS I

DATE: MONDAY 8TH MARCH 2021

TIME: 8.00 A.M - 10.00 A.M

INSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.

2. The following constants may be useful:

$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $g = 9.81 \text{ m/s}^2$ $R = 8.31 \text{ J/moleK}$

Coefficient of linear expansion of steel = $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Young's modulus of steel is = $2.1 \times 10^6 \text{ Nm}^{-2}$.

Wien's constant = $2.90 \times 10^{-3} \text{ m.K}$

- Q1. (a) (i) What is meant by impulse of a force? (1 mark)
- (ii) A golf ball of mass 50 g is struck with a club. The launch angle is 45° , the range is 200 m and the distance the ball travels while in contact with the club is 2 cm. Find the impulse due to the collision. (3 marks)
- (b) (i) What is meant by Newton's second law? (1 mark)
- (ii) Two blocks of mass 100 g and 50 g are placed in contact with each other on a smooth horizontal surface. A horizontal force of 20 N is applied to the larger mass. Find the contact force between the two blocks. (3 marks)
- (c) (i) What is meant by centripetal force? (1 mark)
- (ii) Show that if a body is orbiting with a radius r and velocity v its centripetal acceleration is
- $$a = \frac{v^2}{r} \quad (3 \text{ marks})$$
- (d) (i) What is meant by simple harmonic motion? (1 mark)

- (ii) Show that, for small angle approximation, a pendulum of length L has a period given by

$$T = 2\pi \sqrt{\frac{L}{g}} \quad (3 \text{ marks})$$

- (e) (i) What is meant by Stoke's law? (1 mark)
- (ii) A sphere of radius 0.3 cm drops in oil at 0.5 cm/s. If the densities of oil and iron are 7.8 and 0.94 g/cm³, find the viscosity of oil. (3 marks).

- (f) (i) In words, what is meant Stefan's law? (1 mark)
- (ii) The average radiated power from the sun reaching the earth's surface is 1.395 kW/m². The radius of the sun is 7×10^8 m and the distance from the sun to the earth is 1.5×10^{11} m. Find the temperature of the sun's surface. (3 marks)

- (g) (i) What is meant by area expansivity? (1 mark)
- (ii) A metal sheet of radius 20 cm is heated from 25 °C to 80 °C. If the change in area is 2 cm² find its area expansivity. (3 marks)

- (h) Differentiate between adiabatic and isothermal processes. (2 marks)

- Q2. (a) A rod of length 30 cm has a linear density given by

$$\lambda = 50 + 20x \text{ g/m}^2$$

where x is in m.

- (i) What is the mass of the rod? (4 marks)
- (ii) How far from the $x = 0$ end is its center of mass? (4 marks)

- (b) Find the center of mass of the shape shown in Fig. 1. Lengths are in cm.

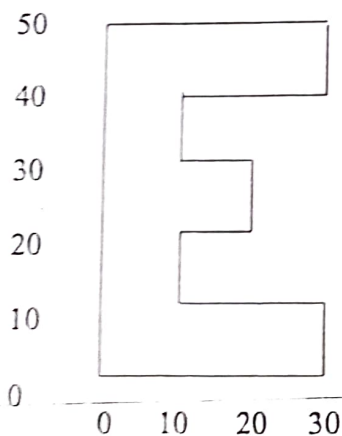


Fig. 1

(12 marks)

- Q3. (a) Show that for an ideal gas in a quasistatic adiabatic process
- (i) $PV^\gamma = \text{constant}$. (4 marks)
 - (ii) $TV^{\gamma-1} = \text{constant}$. (4 marks)
 - (iii) $TP^{(1-\gamma)/\gamma} = \text{constant}$. (4 marks)

- (b) A horizontal spring is found to be stretched 7.5 cm from its equilibrium position when a force of 0.34 N acts on it. Then a 1.125 kg body is attached to the end of the spring and is pulled 10 cm along the horizontal frictionless table from the equilibrium position. What is the
- (i) force exerted by the spring on the 1.125 kg body just before it is released? (2 marks)
 - (ii) period of oscillation after release. (2 marks)
 - (iii) maximum speed of the vibrating body (2 marks)
 - (iv) maximum acceleration of the body (2 marks)

- Q4. (a) Show that for an elastic collision
- (i) the relative velocity of the two objects before the collision equals the negative of the relative velocity after the collision. (3 marks)
 - (ii) if $m_1 = m_2$, the particles exchange velocities. (3 marks)
 - (iii) if m_1 very large compared with m_2 which is initially at rest, the final velocity of m_1 is equal to its initial velocity and the final velocity of m_2 is equal to twice the initial velocity of m_1 . (2 marks)
 - (iv) if m_2 very large compared with m_1 , and m_2 which is initially at rest, the final velocity of m_1 is equal to the negative of its initial velocity and the final velocity of m_2 is zero. (2 marks)

- (b) A proton collides elastically with another proton initially at rest as shown in Fig. 2.

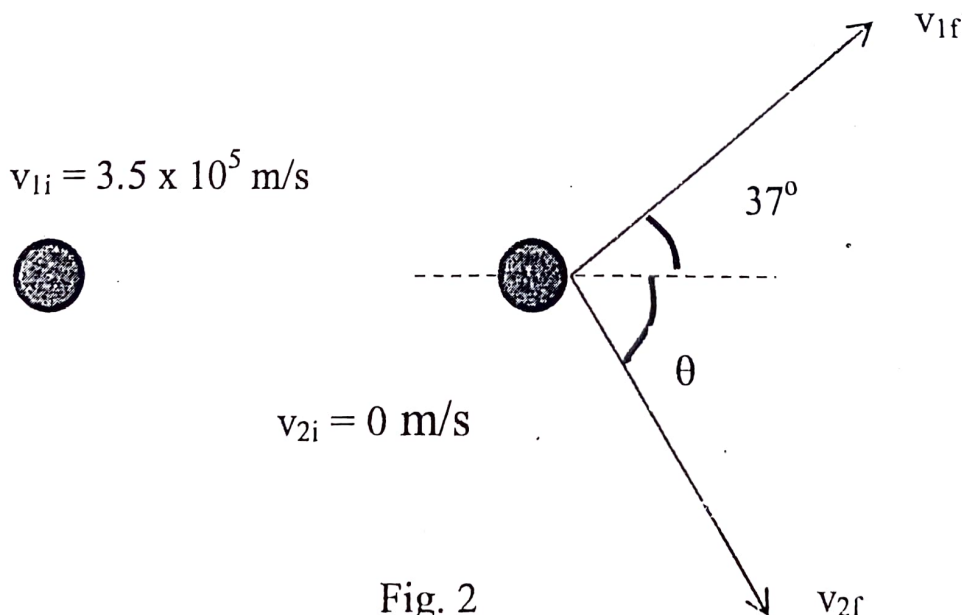


Fig. 2

- | | | | |
|------|------|----------|-----------|
| Find | (i) | θ | (4 marks) |
| | (ii) | v_{1f} | (3 marks) |
| | (ii) | v_{2f} | (3 marks) |

- Q5. (a) Using kinetic theory show that the
 (i) total pressure of molecules against cube walls is given by

$$P = \frac{Nm\langle v^2 \rangle}{3V} \quad (4 \text{ marks})$$

- (ii) internal energy molecules is given by

$$U = \frac{Nm\langle v^2 \rangle}{2} \quad (3 \text{ marks})$$

- (iii) temperature of the system is given by

$$T = \frac{2N\langle K \rangle}{3nR} \quad (3 \text{ marks})$$

- (b) One mole of helium gas has an internal energy of 3600 J and is occupying a cube of side 0.5 m. Find the

- | | | |
|-------|---|-----------|
| (i) | pressure of the gas | (2 marks) |
| (ii) | temperature of the gas | (2 marks) |
| (iii) | x-component of velocity | (3 marks) |
| (iv) | number of collisions per second the in the six walls. | (3 marks) |

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